

INDU 211/2X 2025 – Assignment #3

Problem 1 (Chapter 7, Problem #6)

Annual usage of an item is 1,000 units. Ordering cost is \$10.00 per order, and carrying cost is \$0.50 per unit per year. Determine the optimal order quantity for this item. How many orders will be placed during the year? How many days will there be between consecutive orders? (assume 250 working days in a year.)

Problem 2

Based on the product P1 structure shown below and assume that 200 units are needed for P1 in Week 17, complete the MRP calculation for the left side of product P1. The lead time (LT) of SA1, C1, and C2 are 2, 2, and 1 week respectively. Assume 40 units of SA1 are already in stock and that an order of 60 units of C1 will arrive in Week 12.

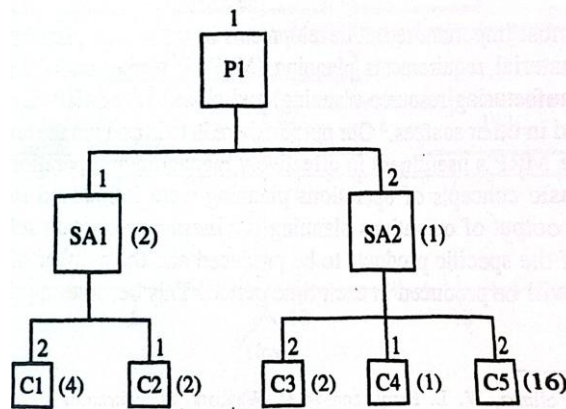


Fig. 7.10. Product structure diagram for product P1.

To complete the question, you may generate a table similar to the one showing in Fig. 7.10 Below.

[illegible]